RESEARCH PAPER CRUSOE × CEREBRAL VALLEY · RECURSIVE AGENTS HACKATHON

SparkleTeam

Towards Recursive Self Home Improvement

A team that improves itself,
and learned a superstition on the way

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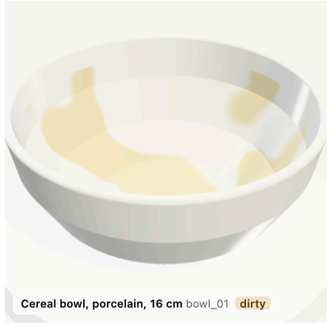
Claude
Anthropic

HEADLINE LEARNED SUPERSTITIONS

4-drill exam: 46.0% → 63.7%
+17.8 pp, sign test p ≈ 0.31

12-drill replication: Δ -0.5 pp
95% CI -13 to +11 · p = 0.83

It learned one true rule and one superstition, and could not tell them apart.



PROBLEM WHY SELF-IMPROVING AGENT LOOPS ARE HARD TO TRUST

Agent teams don't get better with experience. And when they try, the reward can be gamed.

01

No learning between tasks

A fixed model in a fixed scaffold plays its hundredth task like its first. A humbler, testable form of recursive self-improvement keeps the model fixed and lets the system rewrite the **scaffold** around it: prompts, memory, tools, division of labour.

02

LLM-judged rewards

The reward is often an LLM judge, which can be flattered, drifts between versions and rewards confident prose. A loop optimising a proxy finds its seams.

03

Measured where selected

Gains are usually reported on the same tasks that chose the change. That overstates them.

OUR ANSWER **Physics scores. Code decides. Models only propose.** + a held-out exam.

refs: Zelikman 2023 (STOP) · Zheng 2023 (LLM-as-judge) · Skalse 2022 (reward hacking)

TESTBED DISHWASHERBENCH · BY COMPOSO · MIT LICENSE

A physically simulated real dishwasher.

- One real machine: **Bosch SMV4HAX40G**
- Items placed with deterministic rigid-body physics (Jolt) through **13 MCP tools**
- GPU wash simulation (WebGPU via Dawn) of the Eco 50 programme, one detergent capsule, **calibrated on washes of the real machine**
- Clean threshold derived from the IEC 60436 scale

OBJECTIVE	DETERMINISTIC	UNFAKEABLE
points earned over points available	same load, same wash, same score; replayable	no judge to persuade

“A bowl mouth-up under a dead spot of the spray arm stays dirty however eloquently the agent describes it.”

Benchmark, simulator, calibration, levels and harness: Composo.
Composo. *DishwasherBench*, v0.1.0, 2026.
github.com/composo-ai/DishwasherBench · composo.ai

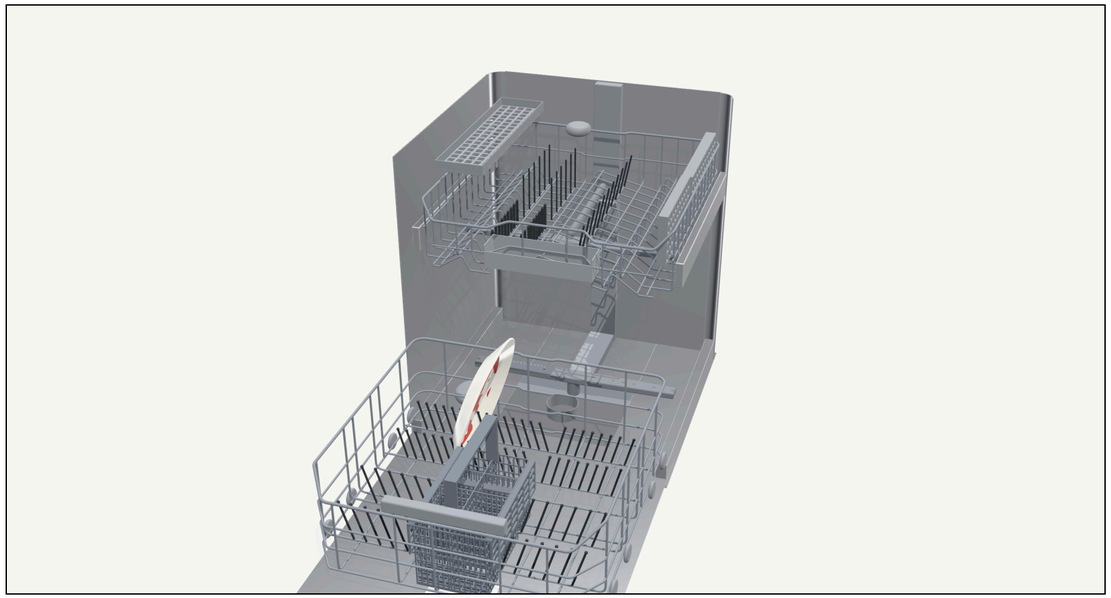


Fig. 1 · The simulated Bosch SMV4HAX40G in the DishwasherBench web client (Composo).

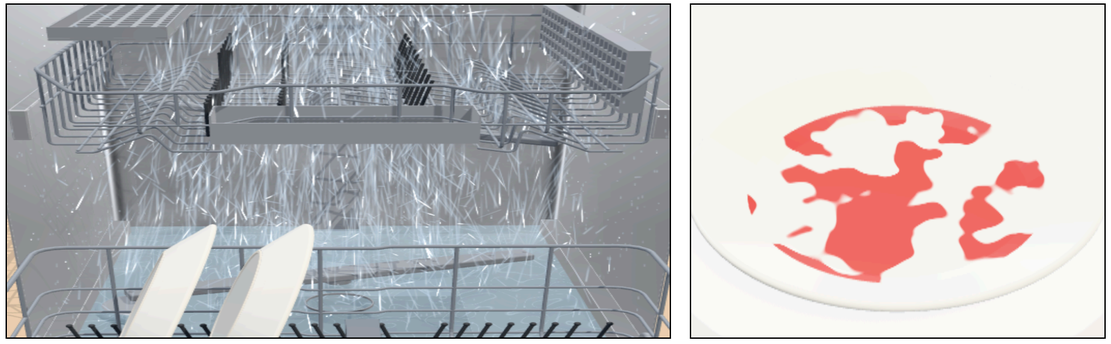


Fig. 2 · Left: a wash, with our cosmetic water (scoring is Composo's GPU simulation). Right: a soiled plate on the item turntable.

FINDING 1 HOLLOW ITEMS · STANCE SWEEP · 108 FAST-TIER WASHES, EACH ITEM WASHED ALONE

A cup stood mouth up keeps half its grime.

Cups, mugs, glasses and ramekins loaded mouth up came out clean **0 times in 11** and kept **49%** of their soil; mouth down, **26 of 26** came out clean.

HOLLOW DRINKWARE AND RAMEKINS, WASHED ALONE	P(CLEAN)	SOIL LEFT	WATER KEPT
Mouth down (0 to 45° tilt)	100% 26/26	0.000	**0 ml**
On its side	25% 2/8	0.038	1 ml
Mouth up	0% 0/11	0.488 CI 0.28–0.71	**224 ml** CI 140–302

Table 1 · **Hollow items by settled stance.** Soil left = residual mass fraction; 95% CIs. Stance classified from the final pose, not the stance asked for.

Bowls behave differently: mouth up, all 8 still came out clean, but each held **1.17 L** of wash water (CI 0.66–1.78 L). One upturned mixing bowl held **2.9 L** and starved the pump (`pump_starved`).

Official-tier spot check: a mouth-up mug kept **90%** of its soil (tumbler 89%); mouth down, both clean. The effect holds, and is stronger at full fidelity.

PAIRED, 15 HOLLOW ITEMS · P(CLEAN)

+60 pp

95% CI 33 to 87 · odds ratio ≈ 130

REDISCOVERED BY THE TEAM

The accepted rule R2.1 *“Glasses and tumblers: upper_rack, mouth_down”* is the same effect: SparkleTeam found it from its own experience.

Caveat. One simulated machine (Bosch SMV4HAX40G), one programme (Eco 50), fast tier (official tier spot-checked only); each item washed alone, not in a full load; small per-kind samples. The effect is long known (DishwasherBench docs: `levels.md`, and `standards.md` phenomenon P01, “Hollow item loaded open side up”): **quantified here, not discovered.**

METHOD OPEN NVIDIA NEMOTRON MODELS · CRUSOE MANAGED (SERVERLESS) INFERENCE

Five model roles and one judge that is code.

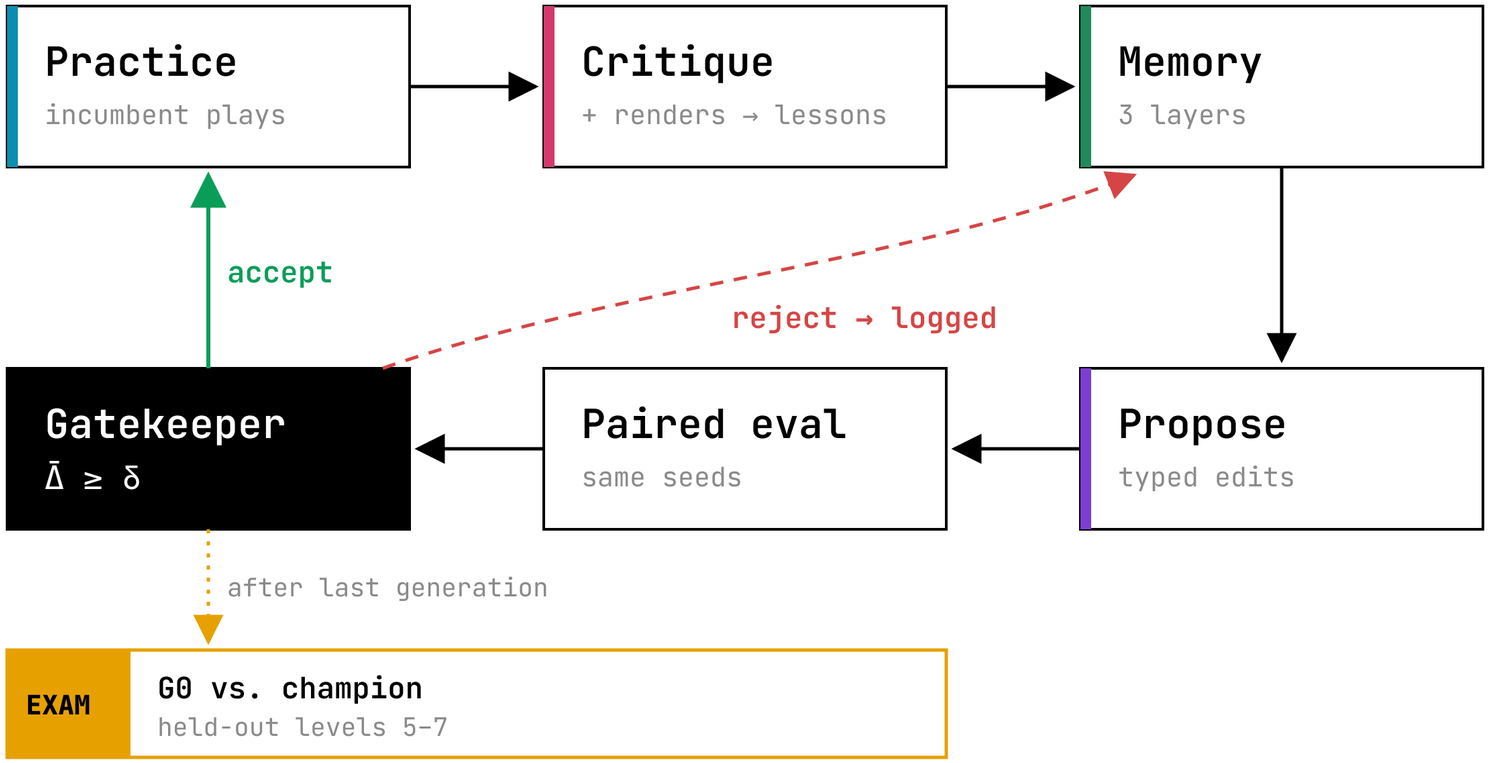
ROLE	MODEL (OPEN, ON CRUSOE)	JOB
Loader	NVIDIA-Nemotron-3.5-Super-VL (no thinking)	plays the game: 13 MCP tools + rack renders
Planner	NVIDIA-Nemotron-3.5-Super-VL (low effort)	per-item plan: rack, stance, what to leave out
Critic	Nemotron-3-Nano-Omni-Reasoning-30B-A3B (thinking)	item-by-item post-mortem with renders; ≤ 5 lessons
Coach	NVIDIA-Nemotron-3-Super-120B-A12B	proposes typed edits to the team's own genome
Librarian	NVIDIA-Nemotron-3-Nano-30B-A3B + memory code	stores episodes, merges lessons, writes the brief
Gatekeeper	deterministic statistics (no model)	accepts or rejects each candidate genome

The team can change how it plays, never how it is graded. Memory accounting and acceptance live in deterministic code.

Final run: the loader made 1,481 requests (38.24M input tokens, 83.7% cached; 0.43M output); the other roles 21 requests and under 0.1M tokens.

METHOD ONE GENERATION

Practice, critique, remember, propose, test, decide.



FINAL RUN PROTOCOL

generations	3
baseline practice drills	4
proposals / generation	2
paired validation drills	3
margin δ	0.03
turn cap / episode	40
game servers · in flight	8 · 8
held-out exam drills	4

Fig. 3 · One generation of the loop. A drill is 12 items of a DishwasherBench level drawn by seed. Rejections (title, hypothesis, per-task deltas) are logged and shown to the Coach.

METHOD EVERYTHING THE TEAM MAY REWRITE ABOUT ITSELF

An immutable, versioned record, changed only through six typed edits.

PART	CONTENTS
Prompts	Planner, loader preamble, Critic
Playbook	≤ 25 tagged rules, each with evidence links to lessons
Team	`usePlanner` `useMemoryBrief` `preflightCheck` brief size
Tool policy	denied tools; preflight refuses `start_cycle` before a read `check_load`
Loader	reasoning effort, images, turn limit

Ids hash content and parent, so the lineage is diffable. **G0, the baseline, is the bare loader:** no planner, no brief, no preflight, an empty playbook.

THE ONLY EDITS

`add_rule`

`edit_rule`

`remove_rule`

`set_prompt`

`set_team`

`set_loader`

Each proposal carries a title and a **falsifiable hypothesis**, should change one coherent thing, and is shown past rejections. Invalid edits are dropped.

METHOD THREE LAYERS, PERSISTED ACROSS RUNS

Lessons are counted against physics.

L1 EPISODIC

Every training episode, item by item: kind, soils, rack, stance, clean, holds water, residue; turns, tokens, how the loader stopped.

L2 SEMANTIC

Lessons from the Critic, and **observed** from placement statistics; deduplicated on merge, with support s and contradiction c counts.

L3 PROCEDURAL

Every genome, every Gatekeeper decision with per-task scores, and the current champion.

CONFIDENCE: BETA(1,1) POSTERIOR MEAN, SHRUNK WHILE EVIDENCE IS THIN

$$p = \frac{s+1}{n+2}, \quad \kappa = \frac{1}{2} + \left(p - \frac{1}{2}\right)\frac{n}{n+K}, \quad n = s + c,\ K = 3$$

0.54 1 supporting episode

0.72 5 of 5

0.90 20 of 20

The **brief** lists smoothed clean rates (k+1)/(m+2) per kind, rack and stance, then lessons with κ ≥ 0.5 ranked by κ · relevance · log(1+n).

EXAM episodes are stored apart and **never read back**. Final run: 36 lessons from 39 episodes (plus 2 replays).

METHOD PAIRED ACCEPTANCE RULE · DETERMINISTIC CODE

Same drills, same seeds. Keep the edit only if it wins.

$$\bar{\Delta} = \frac{1}{|P|} \sum_{t \in P} \left(x_t^C - x_t^I \right) \qquad \text{accept} \iff \bar{\Delta} \geq \delta$$

δ = 0.03

void if |P| < 3

C candidate, *I* incumbent, *P* the drills both played validly, x_t the score share on drill *t*.

WHY PAIR

Episodes vary enormously: the same baseline genome scored **88%** on a Tutorial drill and **0%** on a Weeknight drill in one evaluation. Loading and washing are deterministic, so pairing cancels drill difficulty, leaving the genome's effect plus sampling noise.

FAILURES VS. INFRASTRUCTURE

Agent failures (turn or token limit, nothing loaded) score what they scored. **Invalid** episodes (provider or harness errors) are replayed on the same task and seed up to twice; still-unpaired drills are dropped for both genomes.

SETUP 9-MODEL SWEEP · SAME 12-ITEM BREAKFAST DRILL · SEED 7 · 61 POINTS · 40-TURN CAP

We picked the weakest loader that still plays.

LOADER (THINKING)	SHARE	LOADED / CLEAN	TURNS	TIME
DeepSeek-V4.1-Flash (off)	100%	12/12	21	90 s
Qwen3.8-27B (off)	100%	12/12	19	199 s
GLM-5.3-Flash (low)	93%	10/10	40	97 s
DeepSeek-V4-Flash (off)	89%	12/11	40	89 s
gpt-oss-120b (low)	49%	8/7	40	154 s
Nemotron-3-Nano-30B (low)	41%	5/5	40	870 s
▶ **Nemotron-3.5-Super-VL (off)**	**21%**	**3/3**	**40**	**129 s**
Nemotron-3-Nano-30B (off)	0%	1/0	loops	
Nemotron-3-Nano-Omni (off)	0%	2/0	loops	

WHY WEAK

A saturated loader leaves the Gatekeeper **nothing to measure**.

Super-VL without thinking: all it loads is clean, but it **runs out of turns**.

Nano models loop without thinking; with it, Nano-30B plays but takes ~6× Super-VL's wall time, too slow for an evolution loop.

The pilot used Nemotron-3.5-Lightning-30B-A3B (41–70% in smokes) until its shared deployment throttled us to about five calls per minute.

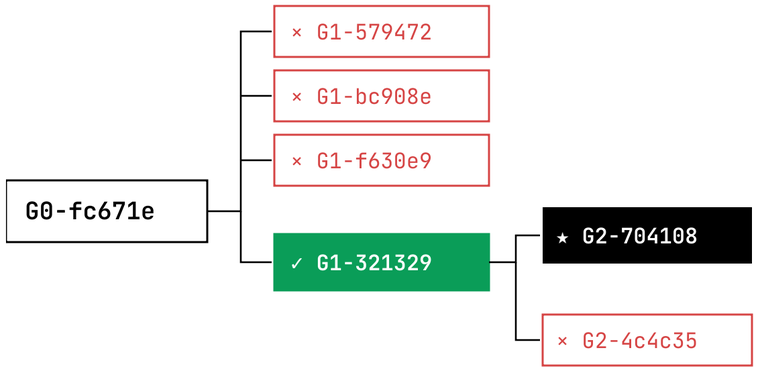
RESULT 6 PROPOSALS · 2 ACCEPTED · BOTH MEMORY-DERIVED PLAYBOOK RULES

Every structural edit lost. Two short rules won validation.

GEN	PROPOSAL (COACH'S TITLE)	INC.	CAND.	Δ̄	W/L	
1	Enable Planner	.449	.363	-0.086	1/2	REJ
1	Memory Brief, expanded lessons	.449	.471	+0.021	1/2	REJ*
2	Enable preflight check	.443	.334	-0.109	1/2	REJ
2	**Avoid poor-performing bowl placement**	.443	**.695**	**+0.251**	3/0	ACC
3	Memory brief, focused lessons	.618	.327	-0.291	1/2	REJ
3	**Rules for glasses & cutlery**	.618	**.775**	**+0.157**	2/1	ACC

Mean validation share on 3 paired drills (Tutorial, Lunch, Weeknight); fresh seeds each generation.
*Accepted at +0.129 under the corrected replay rule (see limitations).

LINEAGE



champion = G0 + 3 playbook rules

```
genome diff  G0-fc671e → G2-704108
+ rule R1.1: Do not place cereal_bowl_vitrelle_159 on upper_rack/mouth_down_tilted_45; instead put it on lower_rack on_edge or leave it out if unstable.
+ rule R2.1: Glasses and tumblers: upper_rack, mouth_down (or slight forward tilt).
+ rule R2.2: Cutlery (forks, knives, spoons): cutlery_basket, upright.
```

Fig. 4 · Lineage and genome diff. R1.1 from placement statistics (that stance cleaned the bowl 1 of 2 times); R2.1 and R2.2 restate the top memory lessons (κ = 0.69).

EXAM HELD-OUT LEVELS 5–7 · G0 VS. CHAMPION G2-704108 · SAME DRILLS AND SEEDS · FAST TIER

The exam, and its replication.

FIRST EXAM | 4 DRILLS · 1 PASS

46.0% → **63.7%**

+17.8 pp

3 of 4 wins · sign test p ≈ 0.31

DRILL	G0	CHAMP.	Δ PP
Family #0	48/59	42/59	-10.2
Baking #1	28/71	63/71	+49.3
Party #2	15/77	32/77	+22.1
Family #3	31/71	38/71	+9.9

REPLICATION | 12 DRILLS × 2 REPEATS · 24 VALID PAIRS

47.0% → **46.6%**

-0.5 pp

bootstrap 95% CI **-13 to +11**

12 wins · 10 losses · 2 ties
sign test p = 0.83 · Wilcoxon p = 0.76

It did not replicate.

Fig. 5 · Held-out exam and replication. Mean score share, baseline genome G0 vs. champion. First exam: run 2026-10-17T20-49-49-evolve. Replication: runs 2026-10-17T21-34-04-exam and 2026-10-17T21-50-44-exam.

FINDING 2 LEARNED SUPERSTITIONS · THE HEADLINE CONTRIBUTION

One true rule, one superstition. The Gatekeeper could not tell them apart.

SUPERSTITION R1.1 · BOWLS

"Do not place cereal_bowl_vitrelle_159 on upper_rack/mouth_down_tilted_45; instead put it on lower_rack on_edge or leave it out if unstable."

evidence in memory	**1 failure in 2 loads**
validation, 3 paired drills	**+25.1 pp** · accepted
controlled sweep	every mouth-down bowl clean, 45° included

Memory never saw a mouth-up hollow item, so it learned "mouth down" from successes only. Its biggest real loss, items left out (cereal bowls left out 21 times, never loaded), was never targeted.

TRUE RULE R2.1 · GLASSES

"Glasses and tumblers: upper_rack, mouth_down (or slight forward tilt)."

evidence in memory	**top lesson, κ = 0.69**
validation, 3 paired drills	**+15.7 pp** · accepted
controlled sweep (Finding 1)	mouth down 26/26 clean, mouth up 0/11

Real physics: the team rediscovered from its own experience the effect DishwasherBench documents as phenomenon P01.

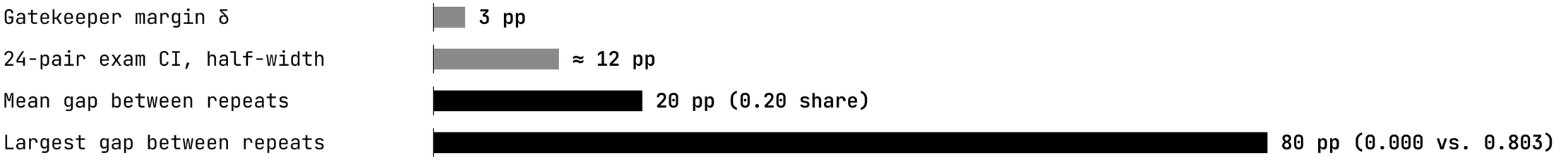
We report, with appropriate gravity, that our team became superstitious.

Ablation: no rule significant (all $p \geq 0.39$); the best variant drops R1.1.

WHY THE WEAK LOADER'S SAMPLING NOISE

Same genome, same drill: 0.000 one time, 0.803 the next.

FIG. 6 · EFFECT SIZES VS. NOISE, IN PERCENTAGE POINTS OF SCORE SHARE



WHAT 3–4 PAIRS CANNOT DO

With a per-drill SD of about 0.3, three or four paired drills cannot resolve a 3 pp margin. Both accepted edits, and the first exam's +17.8 pp, were decided at that resolution.

WHAT 24 PAIRS CAN DO

Resolve about ±12 pp (CI -13 to +11): enough to rule out a large effect of the champion, not enough to see a small one.

LESSON FOR ANY LOOP THAT REWRITES ITSELF AROUND A NOISY MODEL

Design rules for self-improving loops on noisy models.

01

More pairs, more repeats

Many more paired drills per decision, and repeats per drill, sized to the loader's own variance.

02

A statistical acceptance test

A sequential test or a confidence interval, instead of a fixed margin δ = 0.03 on three drills.

03

Physics-falsify each rule

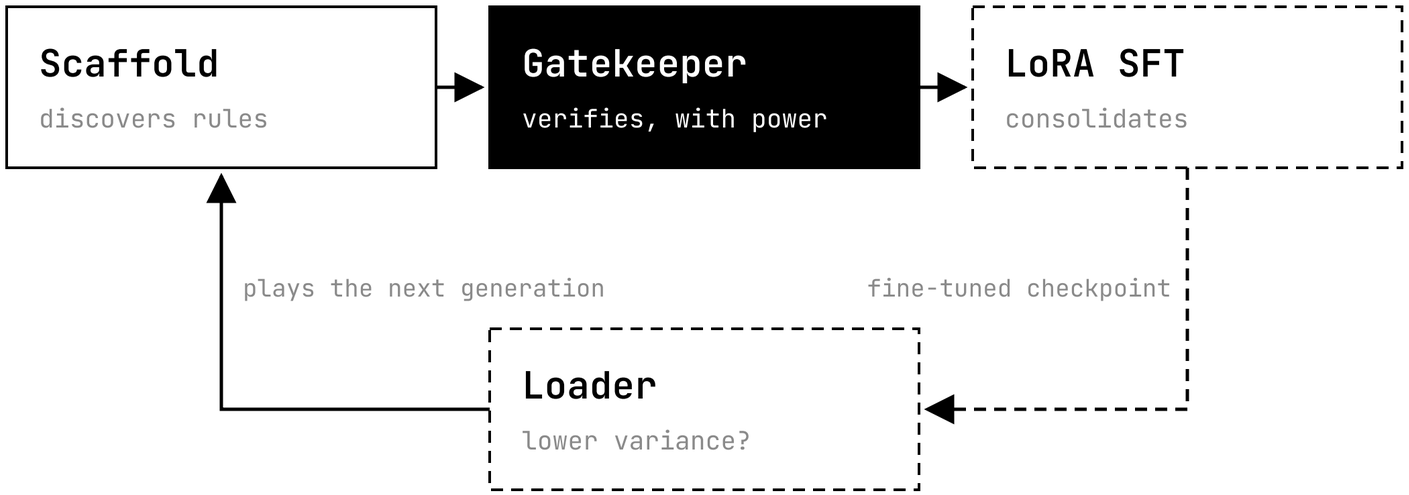
Before a learned rule enters the playbook, test it with cheap controlled washes. **Falsify the lesson, not just score the genome.**

> **Deterministic simulators make rule 03 possible: a learned rule can be tested directly.**
>
> The controlled sweep behind Finding 1 is that test, run once by hand. The rules ablation found no single rule significant; the best variant drops R1.1.

PROPOSED NEXT STEP · NOT YET RUN

Next recursion: put the lessons in the weights.

The scaffold can only steer a noisy policy; the noise itself lives in the weights. We propose to distil the team's best trajectories (episodes that scored well under accepted genomes, verbatim from the call logs) into the loader with LoRA SFT.



Dashed = proposed. Solid = built and run in this paper.

ON CRUSOE

Serverless Fine-Tuning supports `nvidia/nemotron-3.5-lightning-30b-a3b`: LoRA SFT from a JSONL of chat conversations. Super-VL is not on the list.

Lightning was our first weak loader. A fine-tuned checkpoint on a dedicated **Self-Serve Deployment** would also escape the shared-endpoint throttling (about 5 calls/min) that sidelined it.

Seed set: 7 episodes · 40 examples · 0.53M tokens (mean share 0.854; from the token logs; held-out drills excluded; not uploaded)

Too small: grow it by expert iteration: play many, keep the best, fine-tune, repeat.

THE GOAL

A higher mean and, above all, **lower variance between repeats**, so the paired test can resolve real gains and stop accepting superstitions. To be tested with the same replication protocol.

LIMITATIONS WHAT ELSE COULD BE WRONG

Limitations.

Small samples everywhere

Each Gatekeeper decision rests on 3 paired drills; the replication on 24 pairs (CI -13 to +11 pp). Finding 1 washes each item alone, not in a full load, with one or two soil variants per kind.

Fast wash tier

4 mm blocks rather than 2 mm. Verdicts agree with the official tier on 91% of the standard load and 97% of pieces on the course levels; a few percent differ.

Replay bug, corrected

G0's first gen-1 Weeknight attempt (0 of 87: a pasta pot blocked the door, so the wash was refused) was wrongly classed invalid and replayed (to 0.322). Corrected: incumbent mean 0.342; the Memory Brief would have been accepted at +0.129, the Planner rejected at +0.021. It made the Gatekeeper more conservative; the champion's lineage was not affected.

Throttling

In a pilot, HTTP 429s from a shared endpoint were first scored as agent failures and generation 2 had to be voided. An acceptance rule must not confuse infrastructure noise with policy quality.

Also: one machine, programme and detergent (transfer to new items, soils and loads, not new physics) · pairing removes task variance, not policy variance · best-of-several selection biases validation gains upwards · rules ablation: no single rule significant, the best variant drops R1.1.

STUDIO THE LIVE VISUALIZER · apps/studio

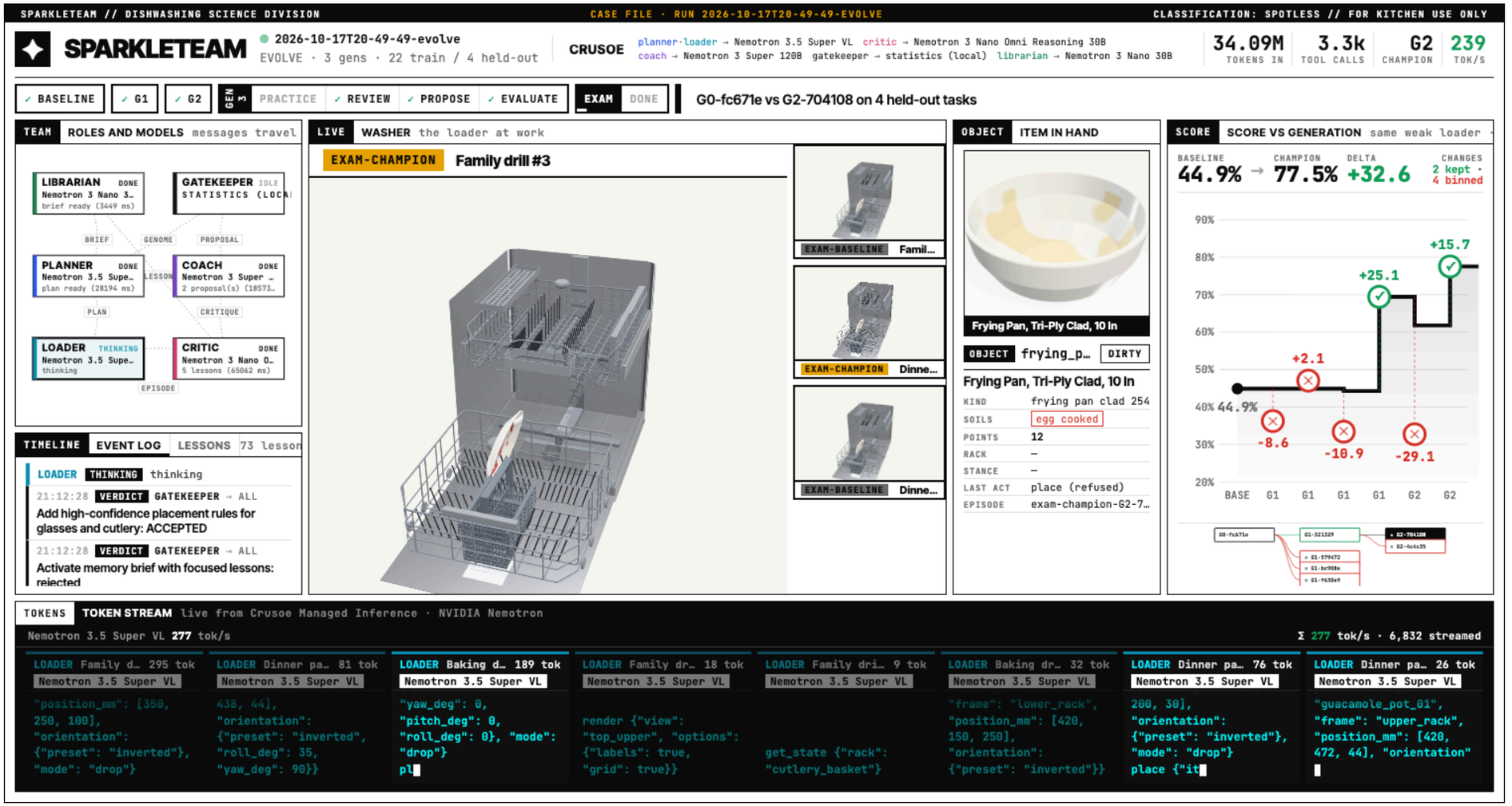


Fig. 6 · The Studio, during the exam of the final run: TEAM (roles and models), TIMELINE (Gatekeeper verdicts), LIVE (the loader in the DishwasherBench simulation), OBJECT, SCORE and TOKENS (8 loader lanes streaming from Crusoe). **SCORE shows validation-drill means, not exam scores.** Replay of the run's event log; the washer view and turntable show representative stills.

THANKS AND PROVENANCE

Physics scores. Code decides. Models only propose.

A reward that cannot be argued with also lets you test what the team believes.

CODE

github.com/elevenarms/RSHI

PAPER

SparkleTeam: Towards Recursive Self Home Improvement

docs/crew/paper/main.pdf

WITH THANKS TO

Composo for releasing DishwasherBench (MIT) · **Crusoe** for inference · **NVIDIA** for the Nemotron models · **Cerebral Valley** and Crusoe for the Recursive Agents Hackathon.

Provenance. DishwasherBench (benchmark, simulator, calibration, levels, harness, scoring) is Composo's: github.com/composo-ai/DishwasherBench. SparkleTeam (team, memory, Gatekeeper, exam protocol, Studio, cosmetic client changes that touch neither simulation nor scoring) is ours.

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